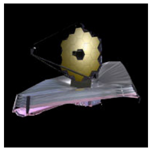

11712 - An Unusual View of the Time Variability of a Rare Exoplanet

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Arthur Daniel Adams (PI)	**The University of Virginia**
Dr. Beth Biller (CoI) (ESA Member)	University of Edinburgh, Institute for Astronomy
Prof. Michael R. Meyer (CoI)	University of Michigan
William Wood Meynardie (CoI)	University of Michigan
Prof. Malena Rice (CoI)	Yale University
Dr. Johanna Vos (CoI) (ESA Member)	Trinity College Dublin
Dr. Niall Whiteford (CoI)	American Museum of Natural History
Dr. Yifan Zhou (CoI)	The University of Virginia

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
HD 106906 b NIRSpec BOTS				
	1	NIRSpec PRISM Time Series	NIRSpec Bright Object Time Series	(1) HD-106906B
HD 106906 b MIRI Imaging TS				
	2	MIRI F1000W Time Series	MIRI Imaging	(1) HD-106906B

ABSTRACT

It is rare for an exoplanet as massive as HD 106906 b to form on the wide orbit we see today. At just 13 Myr of age it orbits completely outside and at a significant angle to its system's debris disk. Adding to its peculiarity, recent findings suggest we are viewing HD 106906 b at the most pole-on view of any L dwarf planet currently known. The atmospheres of worlds like HD 106906 b are expected to be spectrally distinct in latitude, with clearer, bluer poles and cloudier, redder equators. Yet once again, HD 106906 b is an outlier: its near-infrared colors are too red for what is expected

from such an extreme viewing geometry. To add to the ambiguity, while our constraint on the pole-on view relies on measuring the rotational velocity, the current detection of HD 106906 b's rotation is not robust. These points of tension must be resolved in order to properly characterize the dynamical state of HD 106906 b. JWST is uniquely able to resolve both points simultaneously with the wide wavelength coverage and sensitivity of a NIRSpec PRISM time-series observation. With the inclusion of a silicate-sensitive MIRI imaging time series at the same cadence, these observations will allow us not only to probe the physical processes that shape HD 106906 b's atmosphere --- a product of how and where it forms --- but also how they vary across its rotation. The proposed time-resolved observations will expand the sample by being the first spectral variability monitoring both of an L dwarf exoplanet, testing how physical variability mechanisms vary with spectral type; and for a planetary-mass companion around a main-sequence host.

OBSERVING DESCRIPTION

We will use both NIRSpec PRISM BOTS and MIRI imaging time series at F1000W to monitor the variability of the exoplanet HD 106906 b across two rotation periods in each observing mode.

The NIRSpec PRISM BOTS observations will be immediately followed by the MIRI imaging time series.

Each time series covers approximately 8.88 hours, just over twice the rotation period of approximately 4 hours, and each is divided into 35 exposures. Each exposure is approximately 15 minutes. For NIRSpec PRISM BOTS, there are 25 groups per integration and 40 integrations per exposure. For the MIRI imaging, there are 54 groups per integration and 6 integrations per exposure.

Target acquistion for NIRSpec is performed with the SUB2048 subarray and F140X filter.

Proposal 11712 - Targets - An Unusual View of the Time Variability of a Rare Exoplanet

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-106906B	RA: 12 17 52.6100 (184.4692083d) Dec: -55 58 26.60 (-55.97406d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Brown dwarfs, Exoplanet Systems, L dwarfs, Periodic variable stars, Substellar companions]

Observation

Proposal 11712, Observation 1: NIRSpec PRISM Time Series

Fri Mar 13 21:09:43 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Bright Object Time Series

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-106906B	RA: 12 17 52.6100 (184.4692083d) Dec: -55 58 26.60 (-55.97406d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Brown dwarfs, Exoplanet Systems, L dwarfs, Periodic variable stars, Substellar companions]

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	269127.12

Template

Subarray
SUB512

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRS	25	40	35	1	1400	32007.696	269127.14

Special Requirements

Time Series Observation
No Parallel Attachments

Observation

Proposal 11712, Observation 2: MIRI F1000W Time Series

Fri Mar 13 21:09:43 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HD-106906B	RA: 12 17 52.6100 (184.4692083d) Dec: -55 58 26.60 (-55.97406d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Brown dwarfs, Exoplanet Systems, L dwarfs, Periodic variable stars, Substellar companions]

Template

Subarray
FULL

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F1000W	FASTR1	54	6	35	None	1	210	31954.586	269127.23

Special Requirements

Time Series Observation
No Parallel Attachments